Homework 5 Math 210A

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Problem 1. Let $\varphi : G_1 \to G_2$ be a group homomorphism.

(a) Show that if $N_2 \trianglelefteq G_2$, then $\varphi^{-1}(N_2) \trianglelefteq G_1$. Deduce that $\ker(\varphi) \trianglelefteq G_1$.

Proof. Suppose $N_2 \trianglelefteq G_2$. Our goal is to show that $\varphi^{-1}(N_2) \trianglelefteq G_1$; that is, for all $g \in G_1$,

$$g\varphi^{-1}(N_2)g^{-1} \subseteq \varphi^{-1}(N_2). \tag{*}$$

To this end, let $g \in G_1$ be arbitrary. To show (*), let $x \in g\varphi^{-1}(N_2)g^{-1}$ be arbitrary. Then

$$x = g\tilde{x}g^{-1} \text{ for some } \tilde{x} \in \varphi^{-1}(N_2).$$

Since $\tilde{x} \in \varphi^{-1}(N_2)$, it follows that $\varphi(\tilde{x}) \in N_2$. But note that $N_2 \trianglelefteq G_2$ and so for all $g' \in G_2$, we have

$$g'N_2(g')^{-1} = N_2. \tag{†}$$

Since φ is a homomorphism, we see that

$$\varphi(x) = \varphi(g\tilde{x}g^{-1}) = \varphi(g)\varphi(\tilde{x})(\varphi(g))^{-1}$$

Since $\varphi(g) \in N_2$, we have $\varphi(x) \in N_2$ from (†). Thus, $x \in \varphi^{-1}(N_2)$ and so we conclude that

$$g\varphi^{-1}(N_2)g^{-1} \subseteq \varphi^{-1}(N_2)$$

which is our desired result. We see that $\ker(\varphi) \trianglelefteq G_1$ because

$$\begin{aligned}
\ker(\varphi) &= \{g \in G_1 : \varphi(g) = 1_G\varphi^{-1}(N_2)\} \\
&= \{g \in G_1 : gN_2 = \varphi^{-1}(N_2)\} \\
&= \{g \in G_1 : g \in \varphi^{-1}(N_2)\} \\
&= \varphi^{-1}(N_2) \\
&\trianglelefteq G_1.
\end{aligned}$$

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(b) Show that if $N_1 \trianglelefteq G_1$, then $\varphi(N_1) \trianglelefteq \varphi(G_1)$. Deduce that any surjective homomorphism will map normal subgroups in the domain to normal subgroups in the codomain.

Proof. Suppose that $N_1 \trianglelefteq G_1$. To show that $\varphi(N_1) \trianglelefteq \varphi(G_1)$, we need to show that for all $g' \in \varphi(G_1)$,

$$g'\varphi(N_1)(g')^{-1} \subseteq \varphi(N_1). \tag{*}$$

Let $g' \in \varphi(G_1)$. To show (*), let $y \in g'\varphi(N_1)(g')^{-1}$. Then for some $\tilde{y} \in \varphi(N_1)$, we have

$$y = g'\tilde{y}(g')^{-1}.$$

Since $\tilde{y} \in \varphi(N_1)$, then for some $\tilde{x} \in N_1$, we have $\tilde{y} = \varphi(\tilde{x})$. Furthermore, $g' \in \varphi(G_1)$ implies that there exists some $\tilde{g} \in G_1$ such that $g' = \varphi(\tilde{g})$. Since φ is a homomorphism, it follows that

$$\begin{aligned} y = g'\tilde{y}(g')^{-1} &= \varphi(\tilde{g})\varphi(\tilde{x})\varphi(\tilde{g})^{-1} \\ &= \varphi(\tilde{g})\varphi(\tilde{x})\varphi(\tilde{g}^{-1}) \\ &= \varphi(\tilde{g}\tilde{x}\tilde{g}^{-1}). \end{aligned}$$

Notice that $\tilde{g}\tilde{x}\tilde{g}^{-1} \in \tilde{g}N_1\tilde{g}^{-1}$. But since $N_1 \trianglelefteq G_1$, we have for all $g \in G$, $gN_1g^{-1} = N_1$. In particular, $\tilde{g}N_1\tilde{g}^{-1} = N_1$. Hence, $\hat{x} = \tilde{g}\tilde{x}\tilde{g}^{-1} \in N_1$. Hence, we can see that $y \in \varphi(N_1)$ and so we conclude that

$$g'\varphi(N_1)(g')^{-1} \subseteq \varphi(N_1).$$

Hence, we conclude that $\varphi(N_1) \trianglelefteq \varphi(G_1)$. ■

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Problem 2. Let G be a group with $N \trianglelefteq G$.

(a) Suppose $x \in G$ with $|x| = n < \infty$. Show that n divides the order of xN in G/N.

Proof. Our goal is to show that $n \mid |xN|$. Denote $|xN| = k$. By the division algorithm, we can find a unique $q \in \mathbb{Z}$ and $0 \leq r < n$ such that

$$k = nq + r. \tag{*}$$

Since N is a normal subgroup, we know that coset multiplication in G/N is well-defined. Hence, we see that

$$x^k = x^{nq+r} = (x^n)^q \cdot x^r = x^r.$$

Note that $r = 0$ because otherwise if $r \neq 0$ such that $x^r = 1$, then r would contradict the minimality of n. Hence, from (*) we can see that $|x| = n$ must divide k. Thus, n divides the order of xN in G/N. ■

(b) Further show that the order xN in G/N is the smallest positive integer k such that $x^k \in N$.

Proof. Suppose $N \neq \{1_G\}$. Then there exists some $a \neq 0$ such that $x^a \in N$. If $a < 0$, then since N is a group $x^{-a} = (x^a)^{-1} \in N$. This tells us that the set $\mathcal{P} = \{b \in \mathbb{Z}^+ : x^b \in N\}$ is nonempty. Since $\mathcal{P}$ is a subset of $\mathbb{Z}$, the well-ordering property implies that there exists a minimal element d such that $x^d \in N$. In G/N, this corresponds to

$$x^d N = 1_G N.$$

But since the order of xN in G/N is unique, this tells us that $k = d$. Hence, k is the smallest positive integer such that $x^k \in N$. ■

(c) Show that if $G = \langle x \rangle$, then G/N is cyclic.

Proof. Since $G = \langle x \rangle$ and $|x| = n$. Then $|\langle x \rangle| = |G| = n$. Our goal is to show that $\langle xN \rangle = G/N$. From parts (a) and (b), we can see that $|xN| = k$ and k is the smallest such positive integer such that $x^k \in N$; that is, $(xN)^k \in G/N$. Hence, we see that $\langle xN \rangle \leq G/N$. All that remains to show is that $G/N \leq \langle xN \rangle$. Let t be any positive power of xN. We will show that $(xN)^t \in \langle xN \rangle$. Using the division algorithm, we can find a unique $q \in \mathbb{Z}$ and $0 \leq r < k$ such that

$$t = kq + r.$$

Since coset multiplication is well-defined in G/N, we have that

$$\begin{aligned}(xN)^t = x^t N = (x^{kq+r})N = \Big((x^k)^q \cdot x^r\Big)N& \\ = x^r N & \qquad (x^k = 1 \text{ from part (a)}) \\ = (xN)^r. & \qquad (0 \le r < k)\end{aligned}$$

Hence, $(xN)^t \in \langle xN \rangle$ and so we have $\langle xN \rangle = G/N$. ■

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Problem 3. Let G with center $Z(G)$. Recall that $Z(G) \trianglelefteq G$.

(a) Prove that if $G/Z(G)$ is cyclic, then G is abelian.

Proof. Our goal is to show that G is abelian. Suppose $G/Z(G)$ is cyclic. We need to show that

$$xy = yx \quad \forall x, y \in G.$$

To this end, let $x, y \in G$ be arbitrary. Since $G/Z(G)$ is cyclic, we must have that $G/Z(G)$ is abelian. Since $xy \in G$ and $yx \in G$, we have that (because $Z(G) \trianglelefteq G$ and so the coset multiplication is well-defined)

$$(xy)Z(G) = (xZ(G))(yZ(G)) = (yZ(G))(xZ(G)) = (yx)Z(G).$$

This tells us that $(xy)^{-1}(yx) \in Z(G)$ and so we have

$$(xy)^{-1}(yx)g = g(xy)^{-1}(yx) \quad \forall g \in G.$$

In particular, if $g = x^{-1}$, we have

$$\begin{aligned}&(xy)^{-1}(yx)x^{-1} = x^{-1}(xy)^{-1}(yx) \\ &\implies y^{-1}x^{-1}y = x^{-1}y^{-1}x^{-1}yx \\ &\implies x(y^{-1}x^{-1}y) = (y^{-1}x^{-1}y)x.\end{aligned}$$

Since $x \in G$ is arbitrary, we see that $y^{-1}x^{-1}y \in Z(G)$. If $g = y$, then

$$\begin{aligned}&y(y^{-1}x^{-1}y) = (y^{-1}x^{-1}y)y \\ &\implies x^{-1}y = (y^{-1}x^{-1}y)y \\ &\implies x^{-1} = y^{-1}x^{-1}y \\ &\implies xy = yx.\end{aligned}$$

Hence, G must be abelian. ■

(b) Show that if $|G| = pq$ where p and q are primes (not necessarily distinct) then either G is abelian or $Z(G) = \{1_G\}$.

Proof. Suppose $|G| = pq$ where p and q are primes. Suppose $Z(G) \neq \{1_G\}$. Our goal is to show that G is abelian. Since G is a finite group and $Z(G) \leq G$, it follows from Lagrange's Theorem that

$$|Z(G)| \mid |G|.$$

This tells us that $|Z(G)| \in \{1, p, q, pq\}$. If $|Z(G)| = 1$, then we would get a contradiction because we assumed that $Z(G) \neq 1$. If $|Z(G)| = p$, then

$$|G/N| = \frac{|G|}{|Z(G)|} = \frac{pq}{p} = q.$$

Since q is prime, G/N must be a cyclic group of order prime q. From part (a), G must be abelian. Similarly, if $|Z(G)| = q$, then we would get $|G/N| = p$, and so G/N is a cyclic group of order prime p and hence G is an abelian group. If $|Z(G)| = pq = |G|$ and $Z(G) \leq G$, then $Z(G) = G$ and we also get that G is abelian.

If p and q are not distinct, then $|Z(G)| \in \{1, p, p^2\}$. A similar argument will show that G will be an abelian group given that $|G/N|$ is prime. For the case that $|Z(G)| = p^2$, then $Z(G) = G$ and so G is abelian. ■

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Problem 4 (Section 3.2 Problem 5). Let H be a subgroup of G and fix some element $g \in G$.

(a) Prove that gHg^{-1} is a subgroup of G of the same order as H.

Proof. ■

(b) Deduce that if $n \in \mathbb{Z}^+$ and H is the unique subgroup of order n then $H \trianglelefteq G$.

Proof. ■

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Problem 5 (Section 3.2 Problem 8). Prove that if H and K are finite subgroups of G whose orders are relatively prime then $H \cap K = 1$.

Proof. ■

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